Lab 2

Learning outcomes:

1. Able to select appropriate data types to store and manipulate data for analytics
    - Understand how data is stored
    - Difference between numerical and categorical
2. Apply graphical method to discover patterns and anomalies
    - Interpret the statistics (box plot and histogram) for numerical data type
    - Create the box plot in TABLEAU
    - Create the histogram in TABLEAU

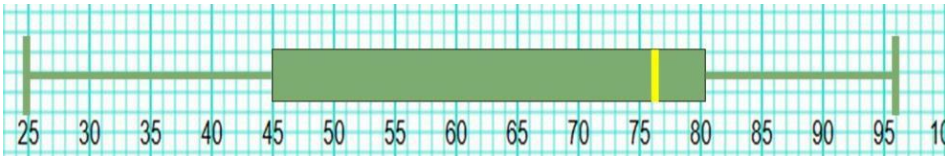
Exercise 1: Creating a box plot

Refer to our last practice, on the subject marks: *Oct21_Marks.xlsx*

Median	**76**
Q1	**44**
Q3	**81.5**
Min	**25**
Max	**96**

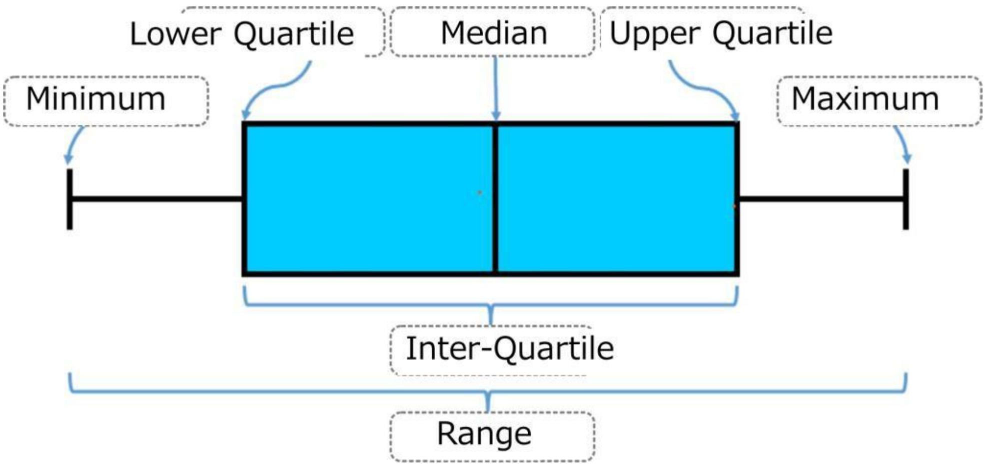
We have determined these 5 numbers, which are important for us to create a box plot.

Draw the box plot using these 5 numbers:



Box plot shows the distribution of data points on the number lines. It gives a better visual of whether the data points are very close or wide apart; whether the spread is wider or narrower above or below median.

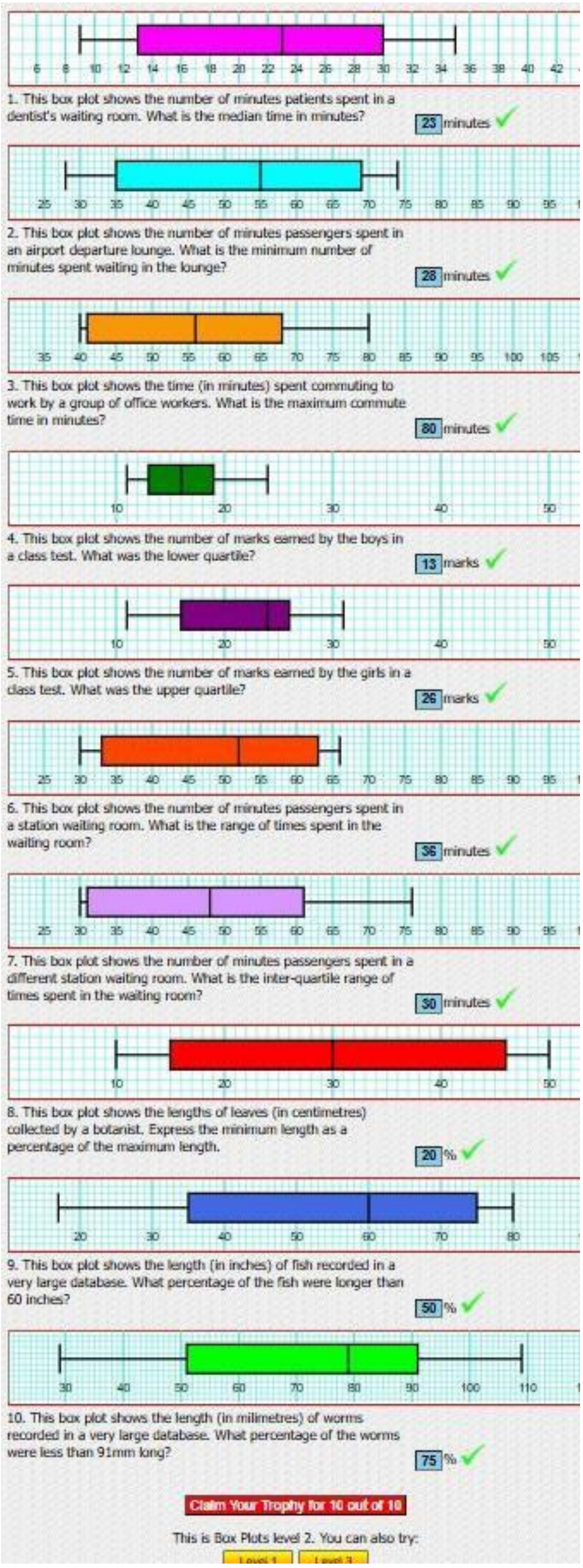
Identify the information found in a box plot



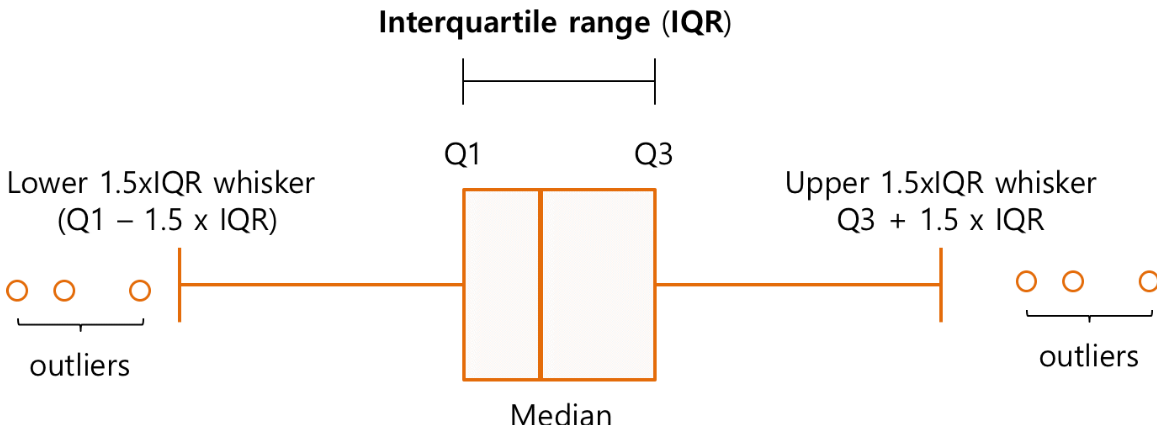
Exercise 2 : Interpreting a box plot

https://www.transum.org/Maths/Exercise/Box_Plots.asp?Level=2

Post a screen shot of at least 3 correct answers.



Exercise 3 : Outliers



https://help.ezbiocloud.net/wp/wp-content/uploads/2018/03/A-typical-box-plot.png

Outliers are data points that are extremely high or extremely low.

How high or how low?

The method used by box plot is

Values that are more than **Q3 + 1.5 * IQR ---------------** (1)

Values that are less than **Q1-1.5*IQR ----------------------** (2)

If there are extremely high or low data points, the "whiskers" are NOT the maximum or minimum value.

The "whiskers" indicate the last data point that is within the calculation range in (1) and (2).

Outliers will appear as "dots" above or below the "whiskers".

Example:

Given Quartile 1 is 30, Median is 37, and Quartile 3 is 44.

Below are the last few data points sort in ascending order:

.... 48, 48, 49, 55, 67, 76, 78, 80 What should be the upper whisker?

What is the value at the upper whisker?

Show your working here

Let's calculate the upper whisker step by step:

1. Find the Interquartile Range (IQR):
$IQR = Q3 - Q1 = 44 - 30 = 14$

2. Calculate 1.5 times the IQR:
$1.5 \times IQR = 1.5 \times 14 = 21$

3. Add this value to Q3 to find the upper whisker limit:
Upper Whisker Limit $= Q3 + 1.5 \times IQR = 44 + 21 = 65$

4. Determine the value at the upper whisker:
The upper whisker should be the highest data point that is less than or equal to 65. In the sorted data:

48,48,49,55,67,76,78,80

The highest data point less than or equal to 65 is **55**.
Therefore, the value at the upper whisker is **55**.

Introduction to TABLEAU

This program explore data with visualisation charts.

Prepare your folder and data

Create a folder on your “desktop” (or document folder) and name this folder “***MyDvis***”.

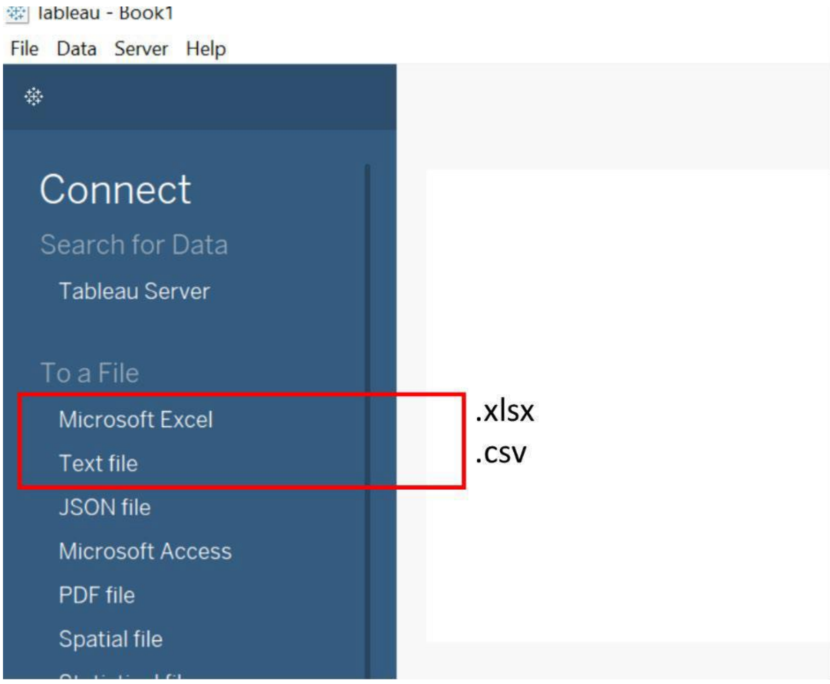
Remember to save all files use in this subject in this folder.

Transfer downloaded data from LMS to this folder (if any).

Exercise 4 – Introduction to Tableau

Launch Tableau.

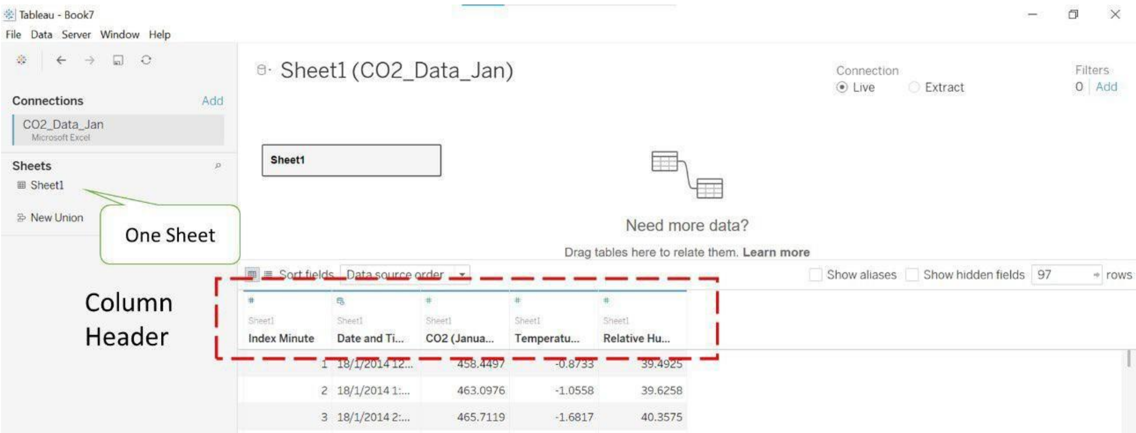
Read File



Depend on the type of file that you want to read.

For this exercise, we are going to read ***CO2_data_Jan.xlsx***

Hence, select "Microsoft Excel", and browse for the data.



This is the first page after loading the file.

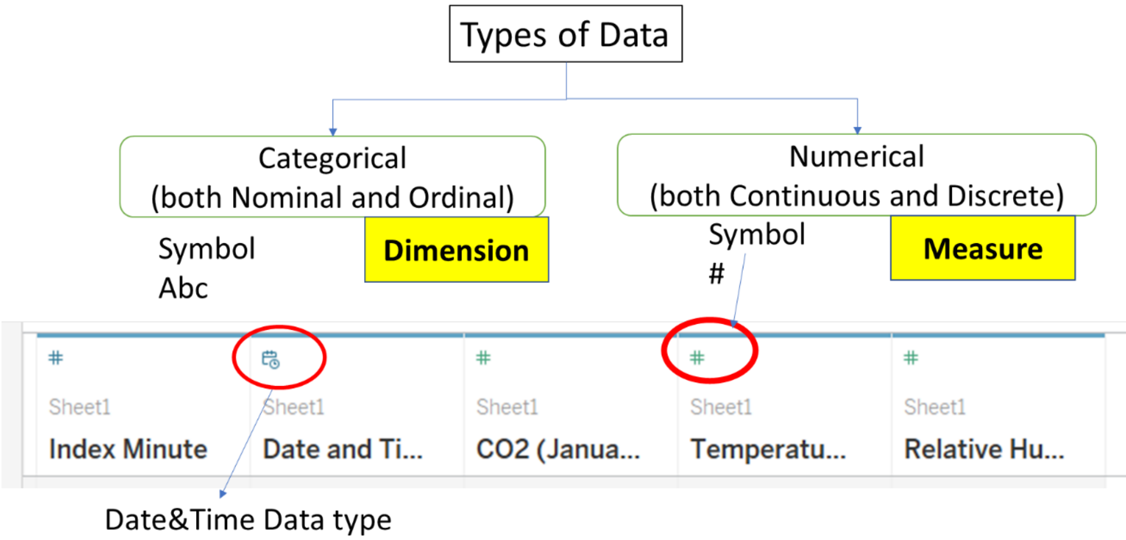
The data is displayed with rows and columns. Each column represents a variable.

Each row is an observation. In this case, each row is a time stamp.

There are 3 Numerical variables, one date&time data and Row Index (per minute increment). There is no categorical type of data.

*(*categorical data refers to text input, example Gender)*

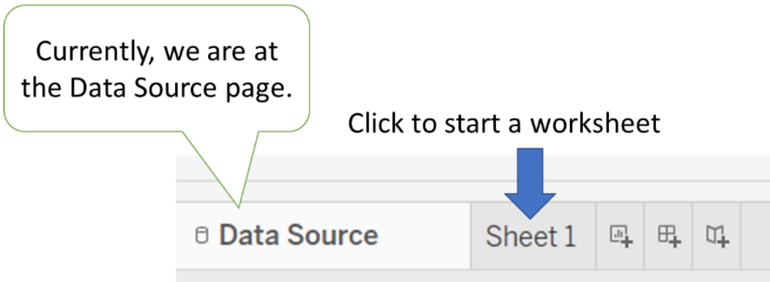
Note the special symbol for each variable in TABLEAU:



Dimension and Measure are the names of the data type used by TABLEAU.

Dimension – represented by symbol **Abc.** For categorical (both nominal and ordinal) types of data

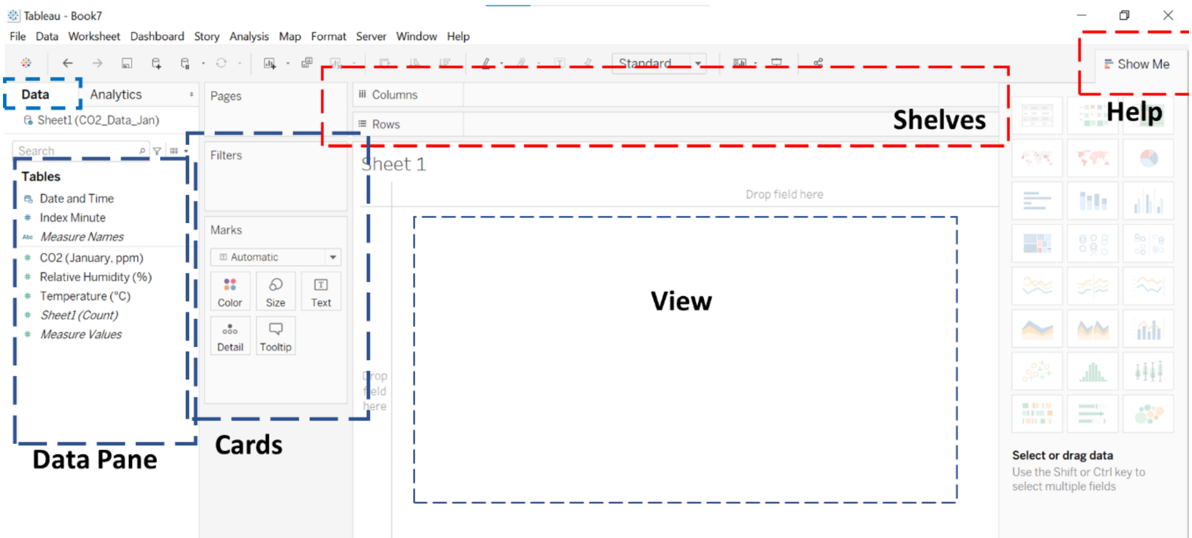
Measure – represented by symbol **#.** For Numerical (both continuous and discrete)



(at the bottom page)

Proceed by starting a new worksheet.

This is the layout in a Sheet.



Note in TABLEAU, there is no workflow. It is just plotting a chart.

Column – can think it as the variable for the X-axis

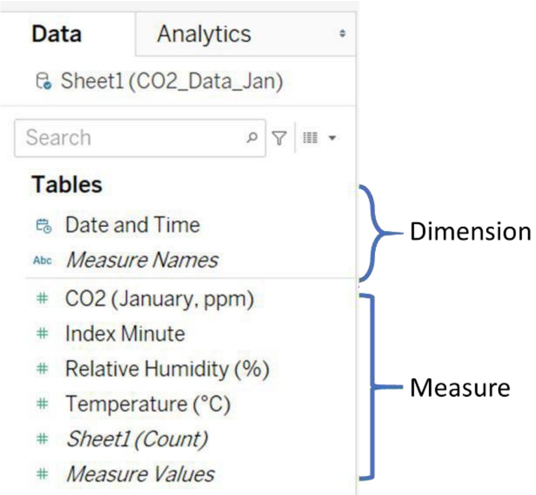
Row – can think it as the values assign to the Y-axis

You can drag and drop any variables in the “Data Pane” to the column and row to create a plot.

The options in the “Cards” pane, are display options. We will use them when we need it.

It is important to understand the arrangement in the Data Pane.

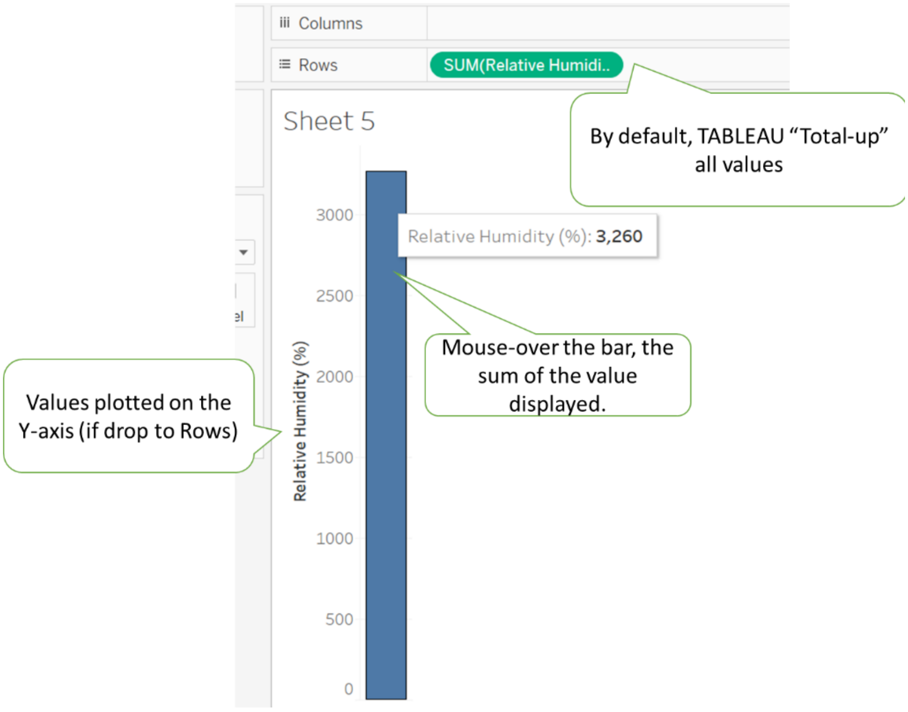
Data Pane



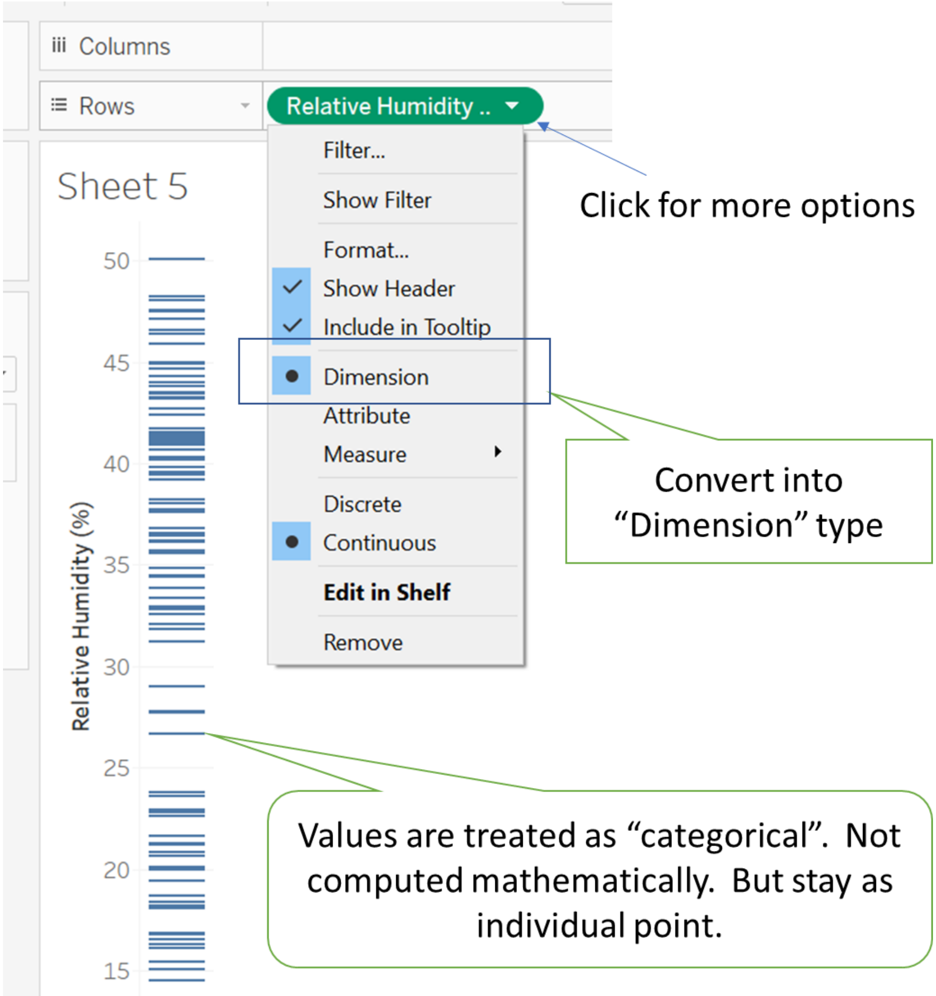
This pane lists all the variables and split the variables into Dimension (categorical data type) and Measure (Numerical data type).

If you drag data from "Dimension" to columns/rows shelve, each value is plotted as individual mark on the axis. In this data set, there is only the special date&time data, no categorical (or you can think it as qualitative data).

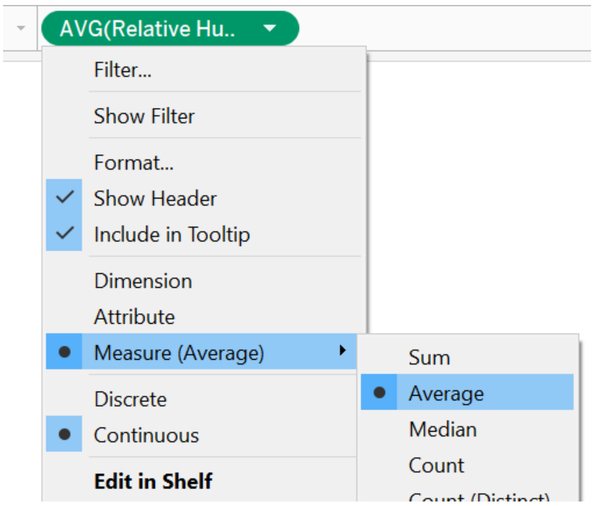
If you drag any data from the "Measure" to columns/rows shelves, values will be plotted along the X or Y-axis (as continuous reading). And display a horizontal or vertical bar showing the SUM of values of that variable. Example:



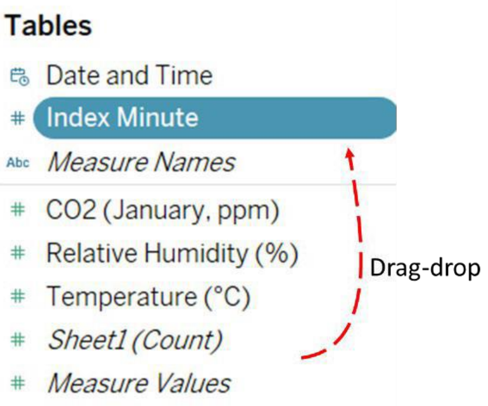
Something interesting about TABLEAU: Data type is easily converted to "Dimension"



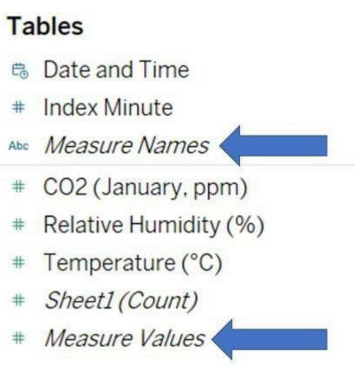
Let's convert back to "Measure", and find the "Average"



Note that "Index Minute" is in the Measure Field. As this data is just an indexing, not meant to be calculated. Drag the variable to the "Dimension" Field. (This is also a way to switch data type)



Note, in TABLEAU, variable is easily change to Dimension by drag-drop.



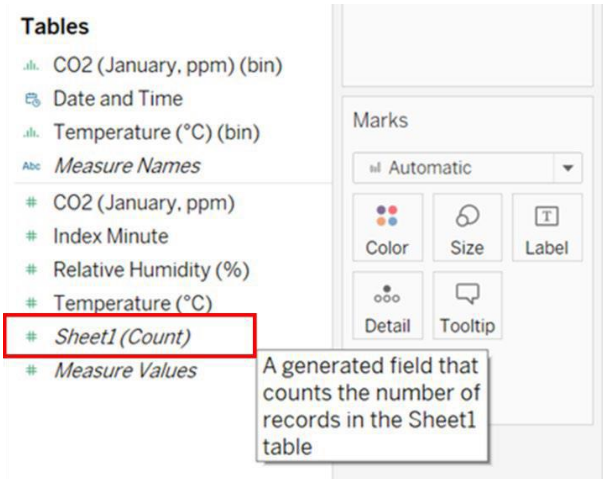
There are two additional variable called "**Measure Names**" and "**Measure Values**". This is generated by TABLEAU. It will show the name of the variable and the value when drop into the view.

If you drag any of these into the "Filter" card, it allows you to manually select. Very flexible!

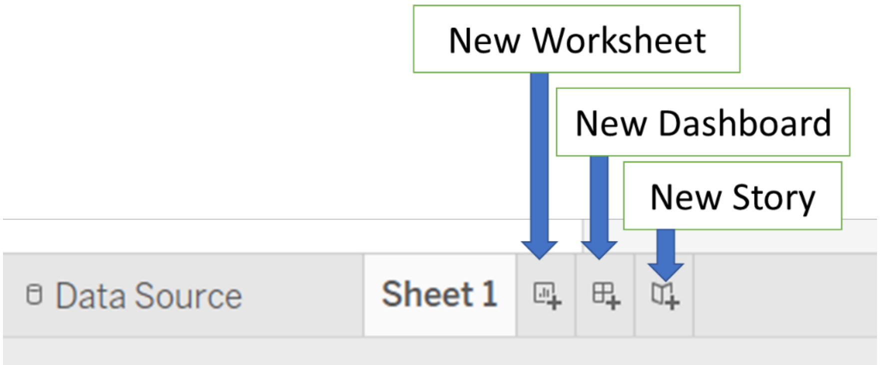
Other from Measure Names and Measure Values, the (Count) variable is also generated by Tableau.

This is used quite frequently.

In this course, we may not be able to cover all the features in TABLEAU. Hence, at times, you need to explore on your own.



There are 3 different options beside the “Data Source” at the lower left corner:



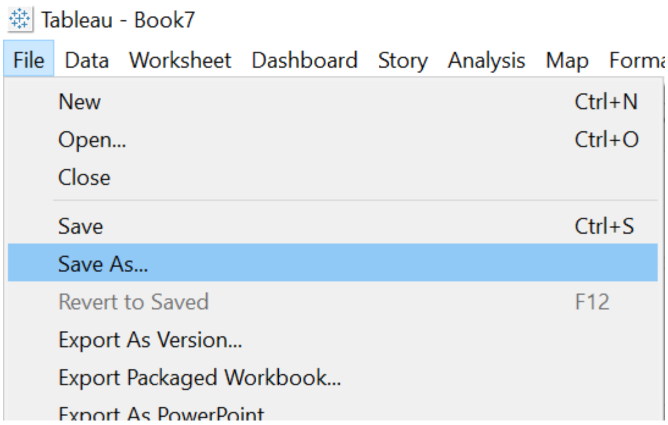
Most of the times, we add “New worksheet”. Every chart is a new worksheet.

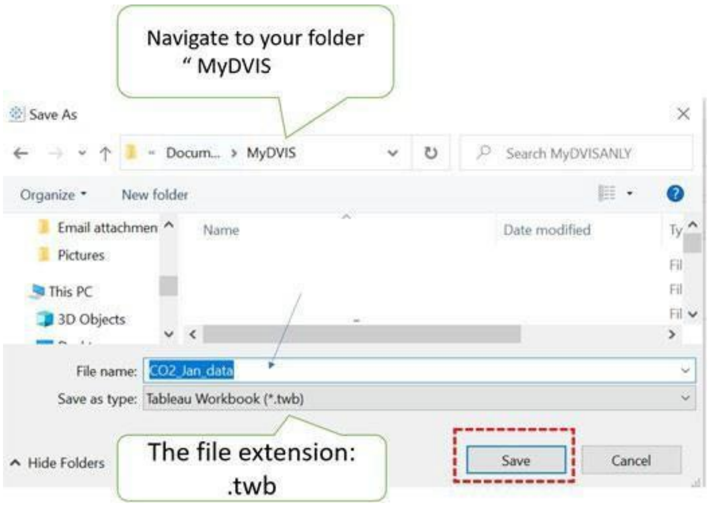
How to save a file?

Before we go further, let’s learn how to save our work in TABLEAU.

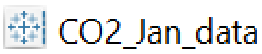
A file in TABLEAU is called a “workbook”. Currently “workbook” is not saved. When you saved a “workbook”, all “Sheets” are automatically saved.

Save the “workbook” as “*CO2_Jan_data*”.



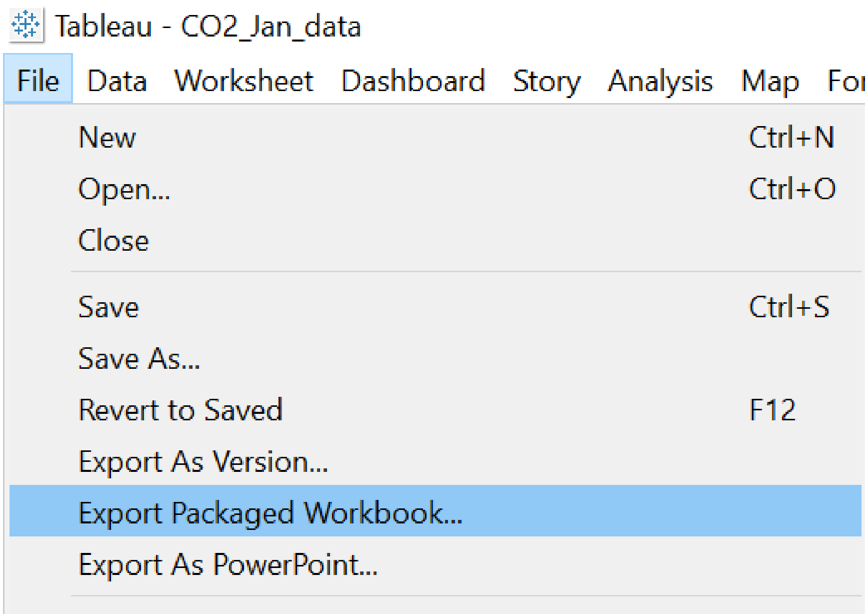


If you open your file explorer, you can find the workbook in your chosen folder:



Note that this workbook does not embed the data file. So, your data file and workbook must be in the same folder, when open this workbook.

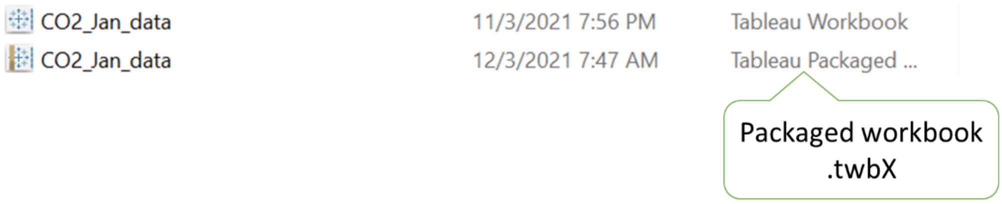
When you submit your work in LMS, you must EXPORT your workbook, so that the data files are embedded,



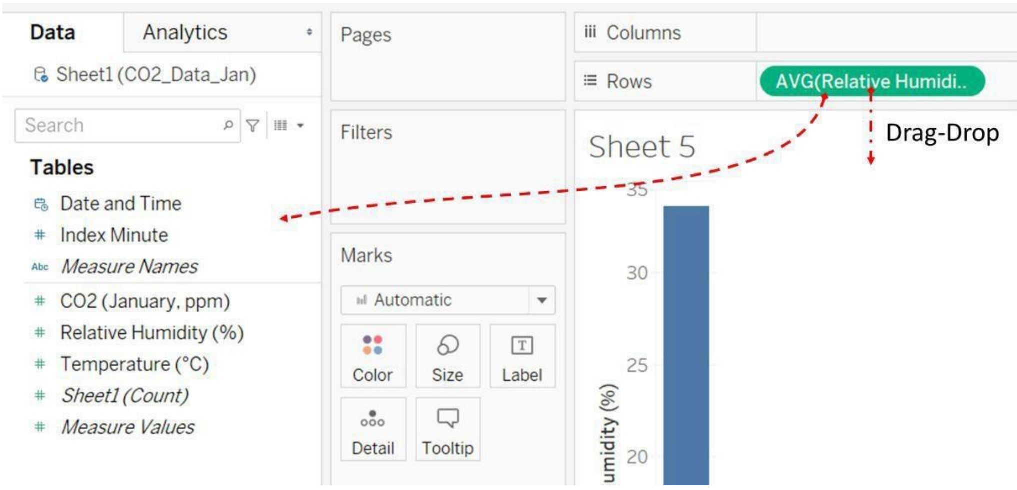
File > Export Packaged Workbook

Important to use this method to save your file for submission.

The file extension of a packaged workbook is **.twbx**

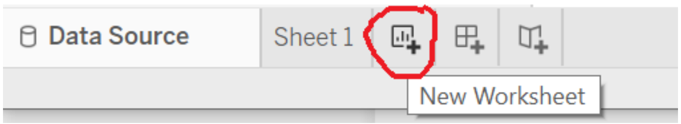


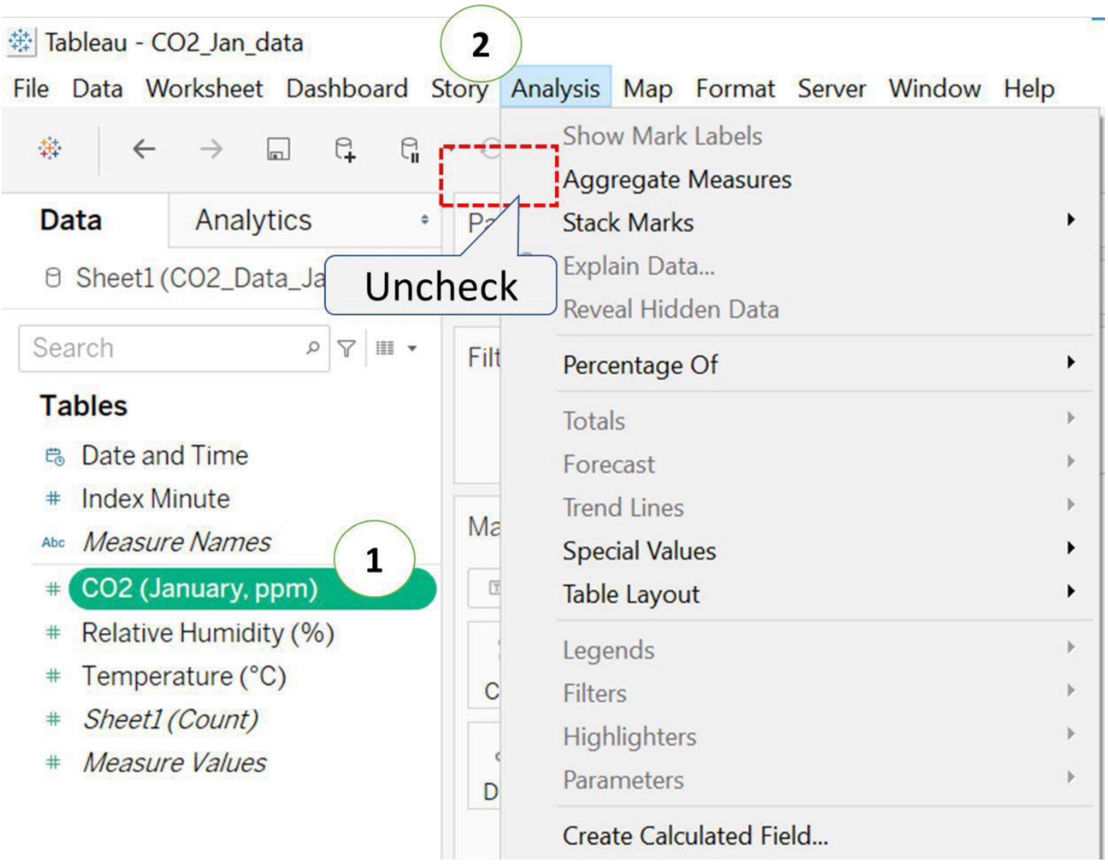
Let’s clear the “shelves”, if you have added anything to it, by dragging the “pills” and drop anywhere (into the View or Data Pane)



<u>Exercise 4A: How to create boxplot?</u>

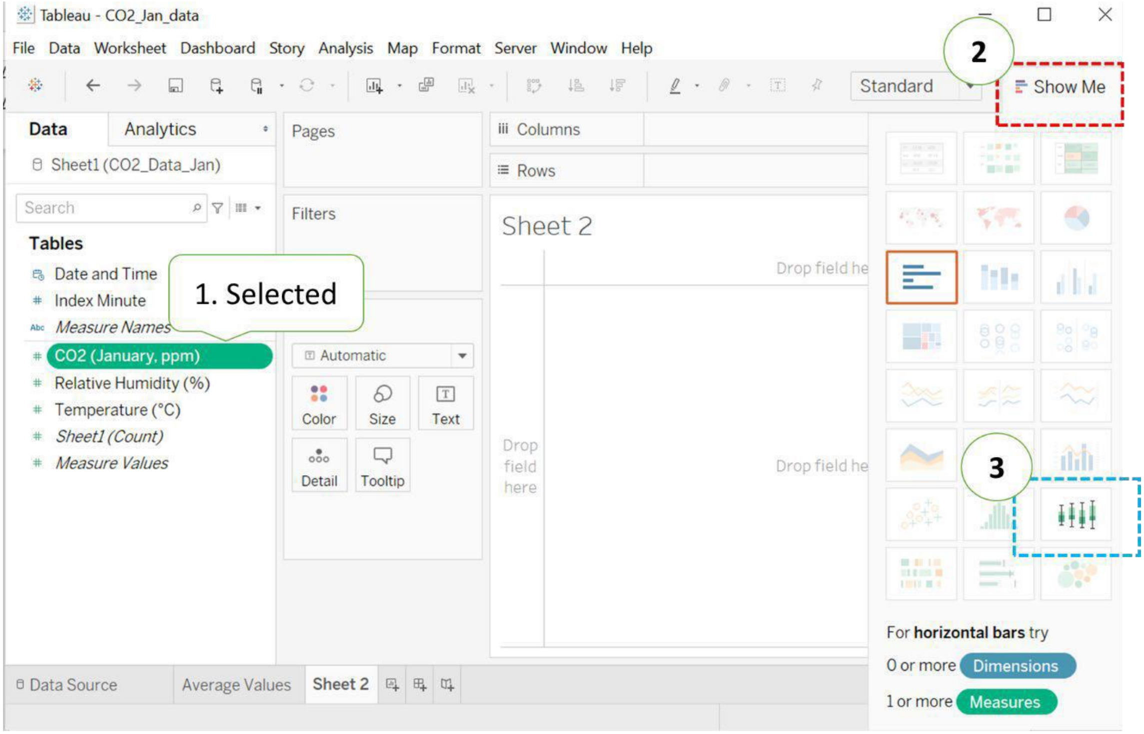
Let's start a new Worksheet.

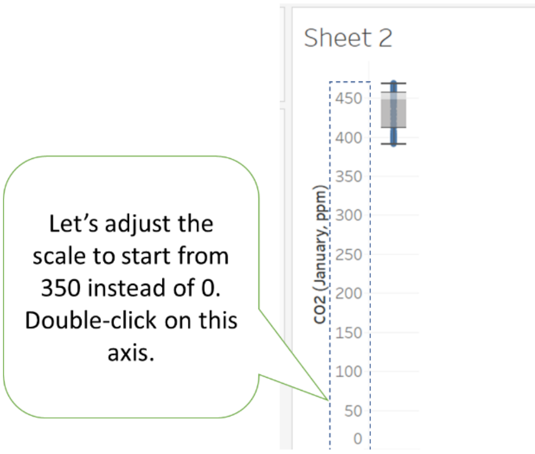




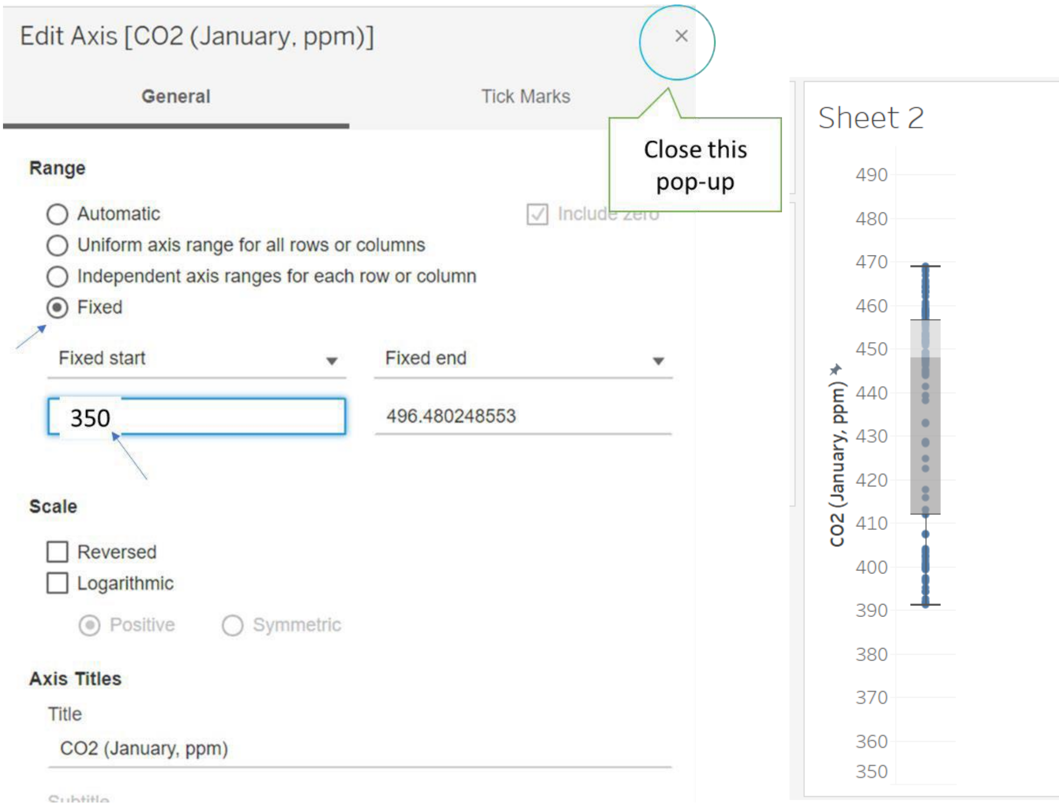
==Very important: uncheck "Aggregate Measure"== if you want to create boxplot.

By default, TABLEAU will always "Aggregate" the values, like add-up and compute the Mean. To create boxplot, this option must "uncheck" so that all points will be plotted.



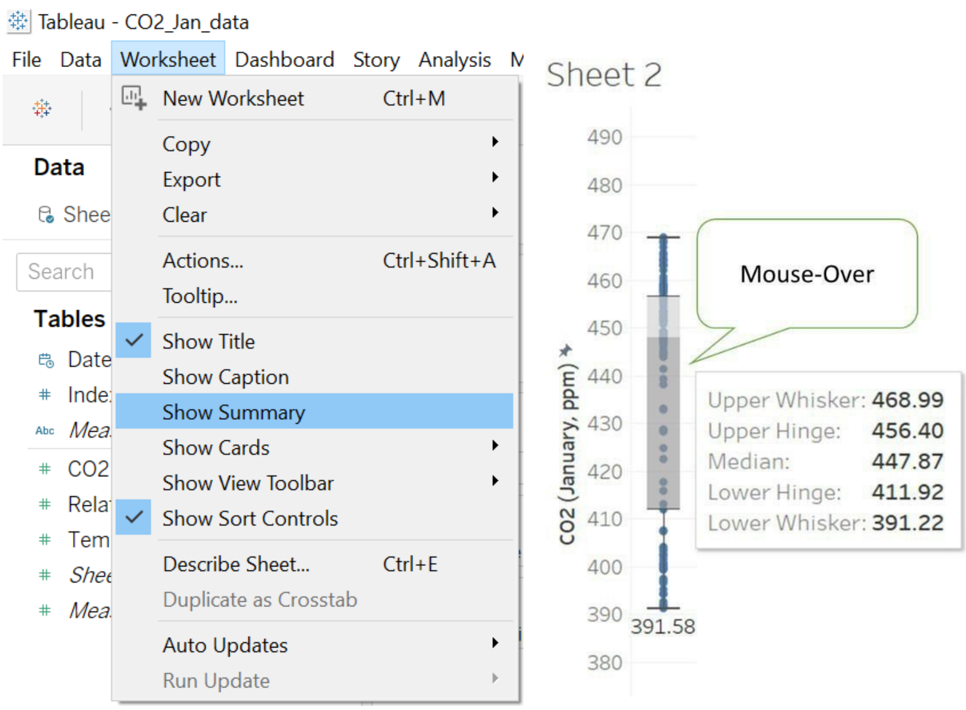


Double-click on the reading pane.



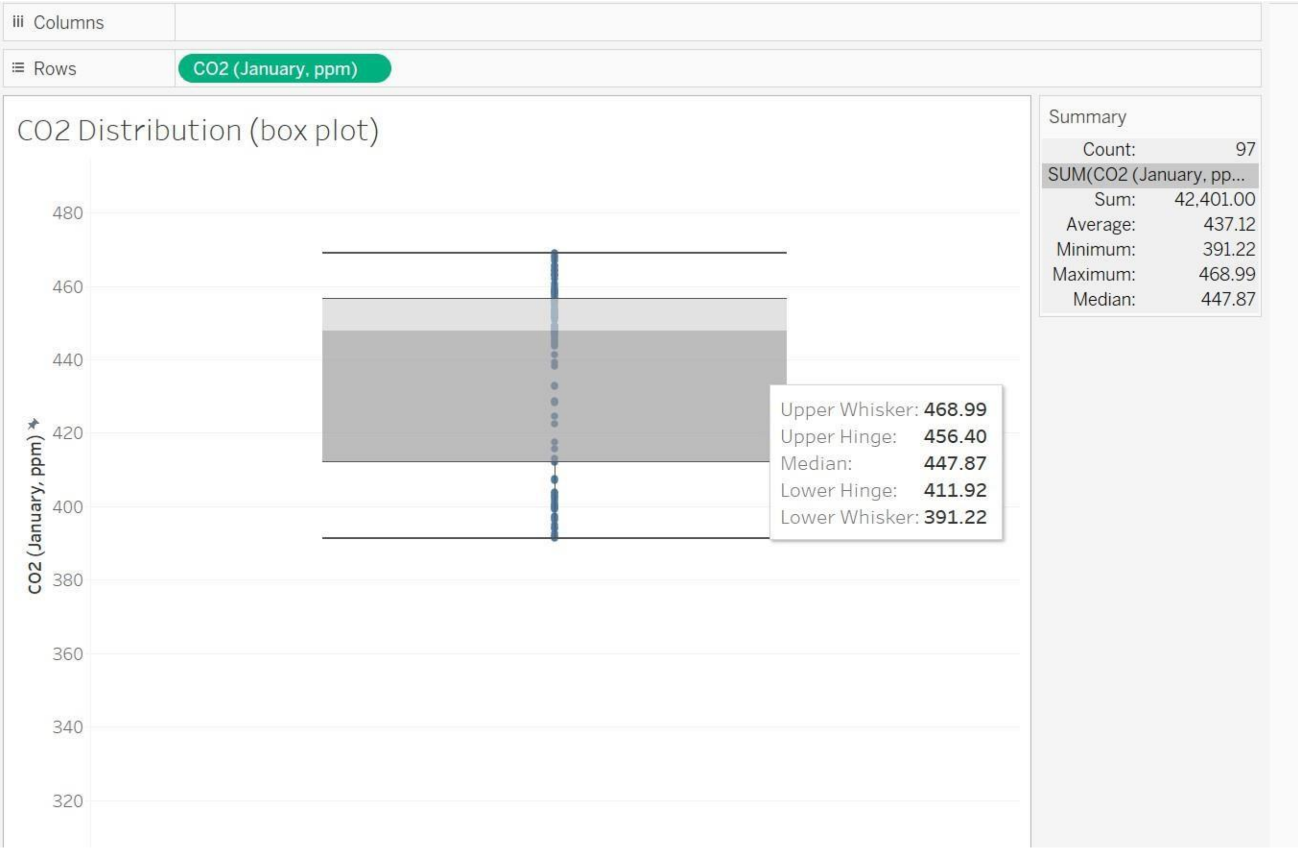
Mouse over the box-plot and you can read the statistics.

Or, top-bar menu: worksheet > Show Summary



The Lower Hinge and Upper Hinge are the Lower Quartile (Q1) and Upper Quartile (Q3) respectively. The Lower Whisker is the minimum and Upper Whisker is the maximum.

Name the worksheet as "CO2 Distribution(boxplot)"

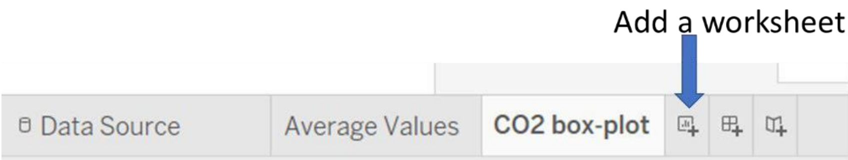


Insights:
The CO_2 level has more variability below the Median, 447.87 ppm (parts per million)
There are no outliers.

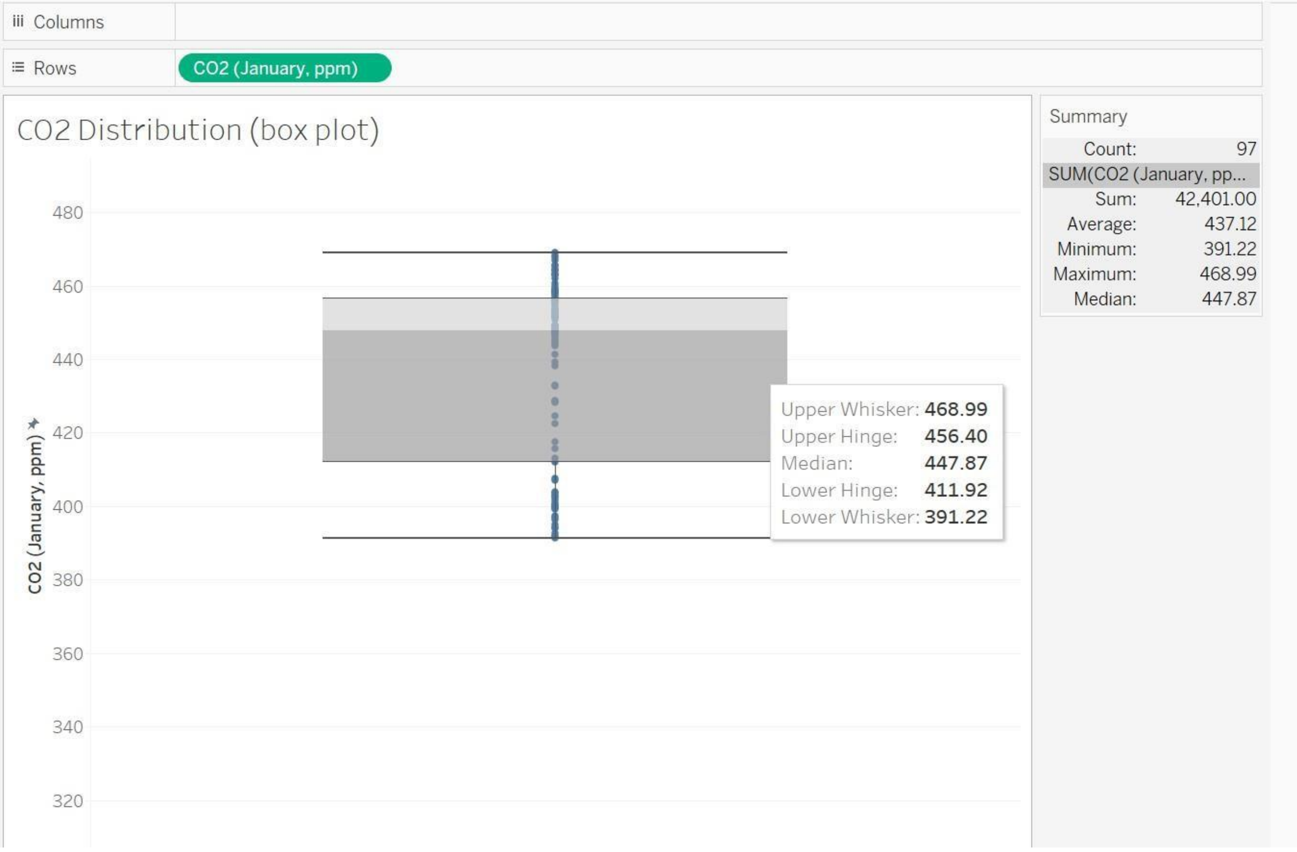
(more insights with comparison)
50% of the times the CO_2 stays below 447.87ppm.
The air quality is acceptable with well-ventilated outdoor air.
Ideal CO_2 leve for outdoor is 350 – 450 ppm
(ref:Acceptable Levels of CO2 in Your Home: What Is Safe? – YourIAQ)

<u>Exercise 4B:</u>

On your own, create boxplot for Temperature:



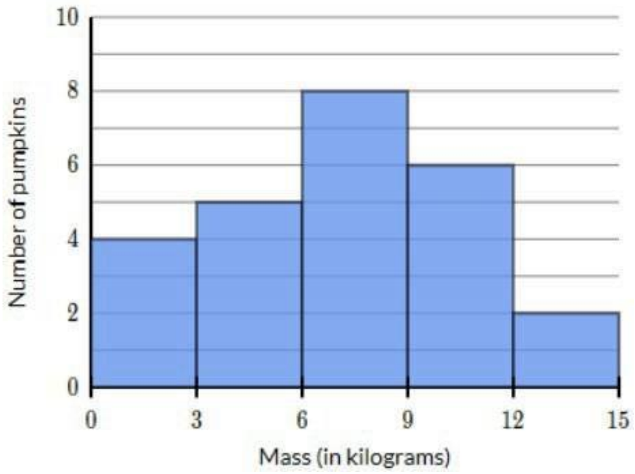
Show your chart here:



What insights can be interpreted from the boxplot?

The CO_2 level has more variability below the Median, 447.87 ppm (parts per million)
There are no outliers.
(more insights with comparison
50% of the times the CO_2 stays below 447.87ppm.
The air quality is acceptable with well-ventilated outdoor air.
Ideal CO_2 leve for outdoor is 350 – 450 ppm
(ref:Acceptable Levels of CO2 in Your Home: What Is Safe? – YourIAQ)

Exercise 5: Insights from a histogram



Mode = 6 to 9 kg

Bin size = 3 kg

Shape of the distribution is Bell-shaped; Single Mode

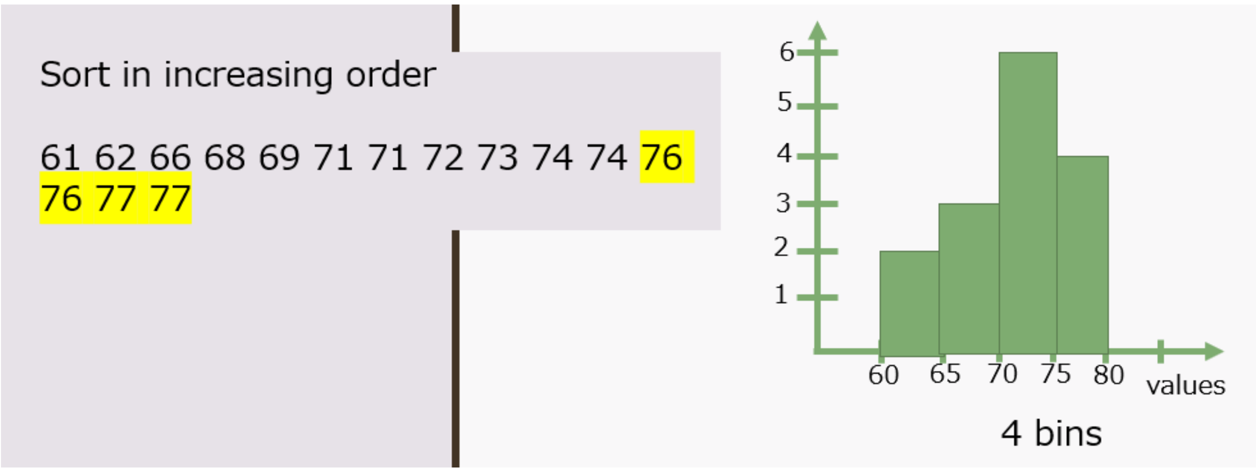
What is the % of pumpkins weigh between 12 to 15 kg? 8%

<u>Exercise 6 – Plot a histogram</u>

Plot the histogram of the following data:

71 ,74, 69, 76, 77, 72, 71, 62, 68, 76, 66, 74 , 77 , 73, 61

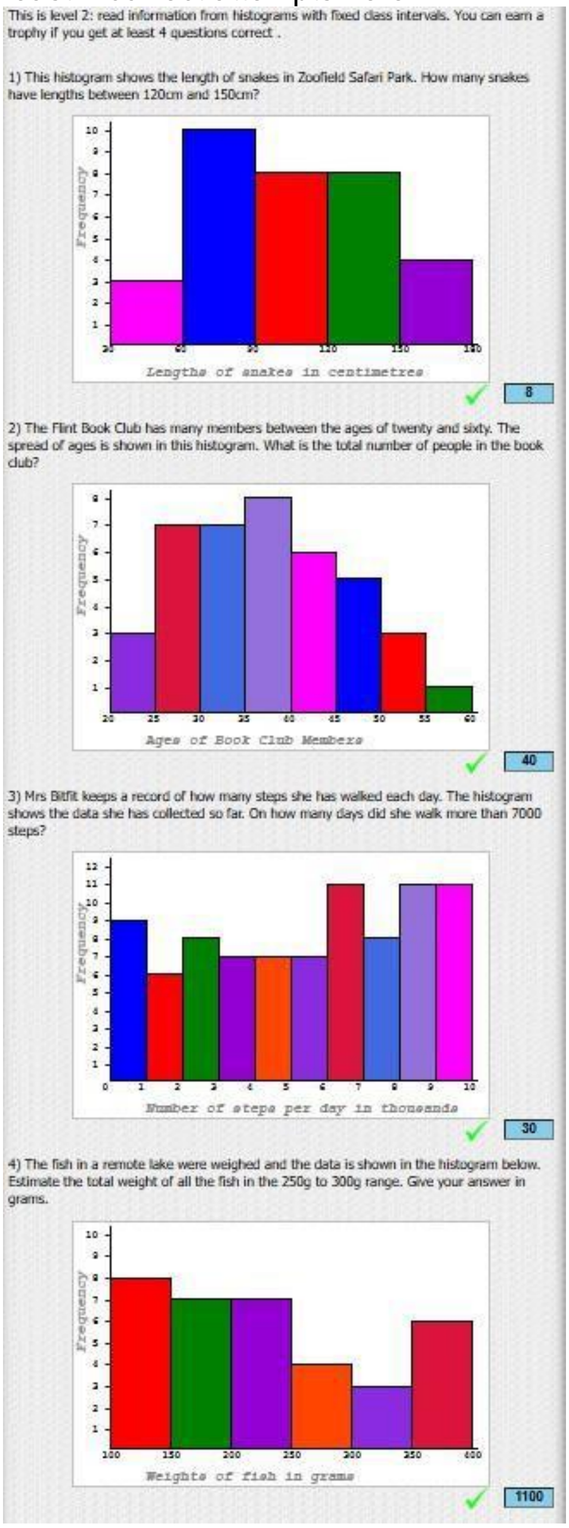
Use constant bin interval of 5.



Exercise 7 – Interpreting a histogram

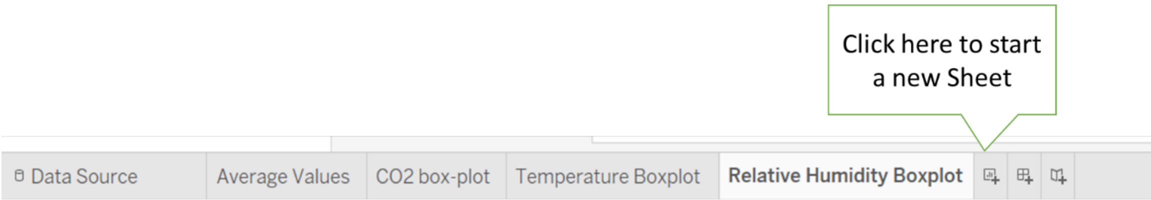
Do the activity in this site:
https://www.transum.org/Maths/Exercise/Histogram/

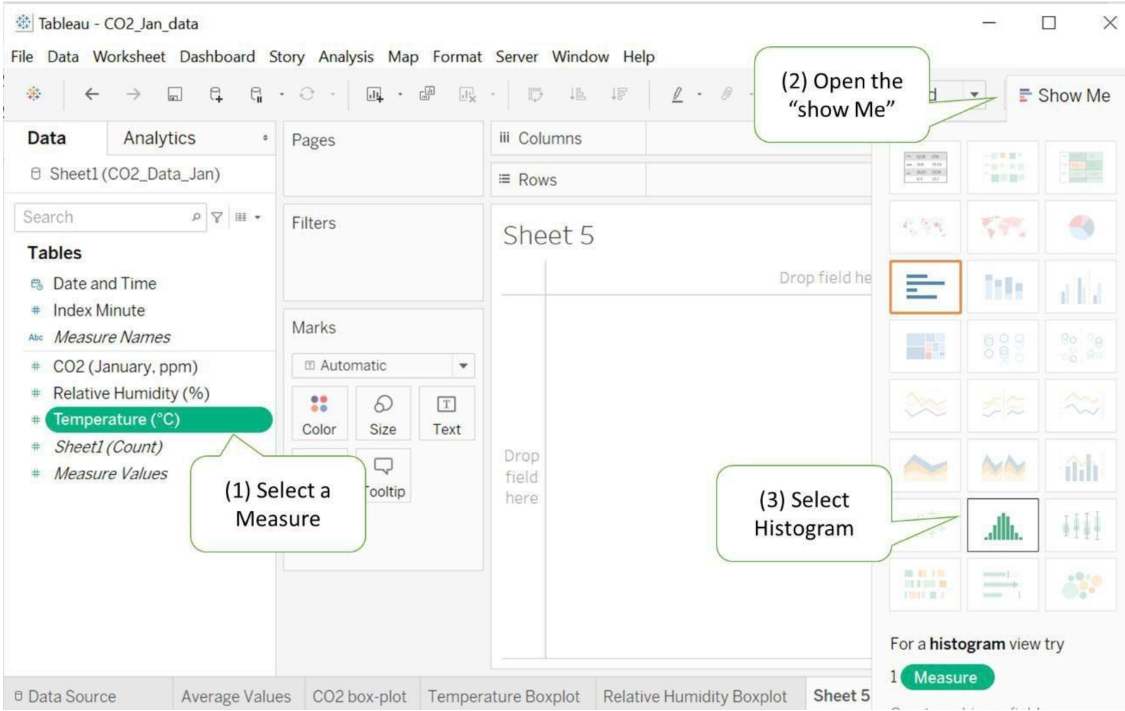
Select Level 2. Post at least 2 correct attempts here.



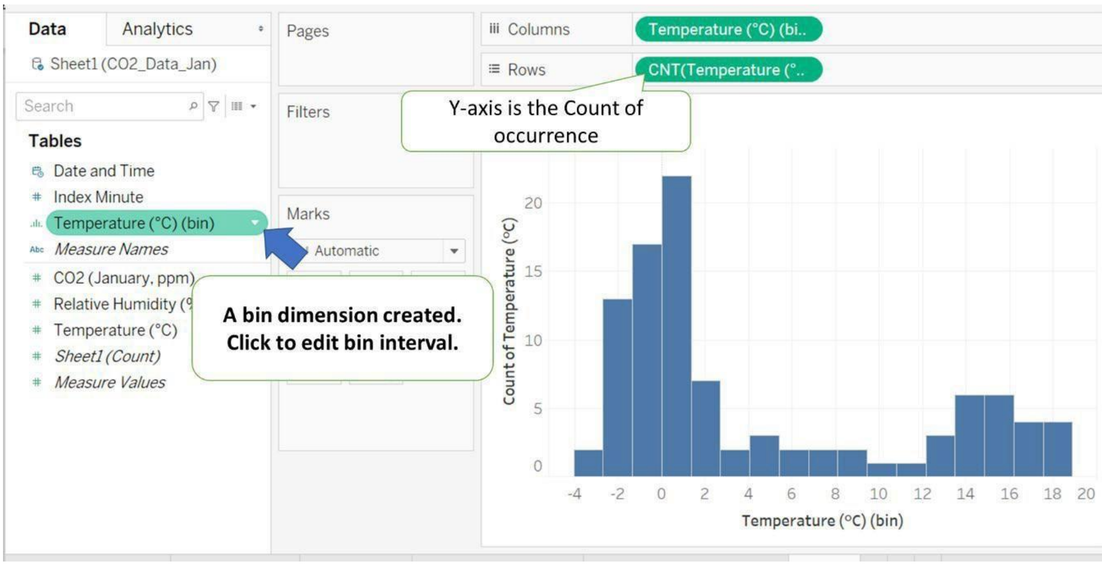
Exercise 8: How to create a histogram?

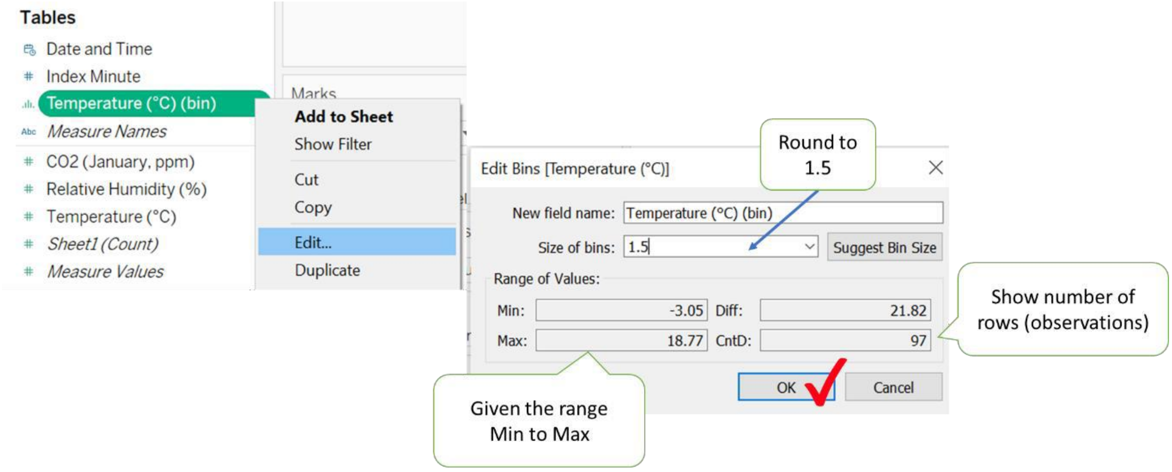
Start a new Sheet.

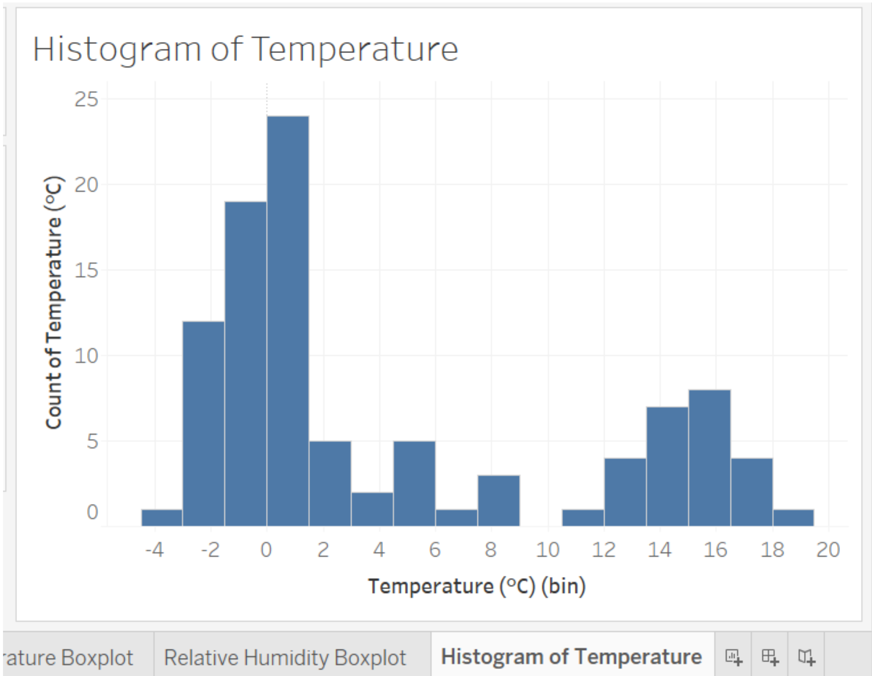




Click on the "Show Me" again to close the list.

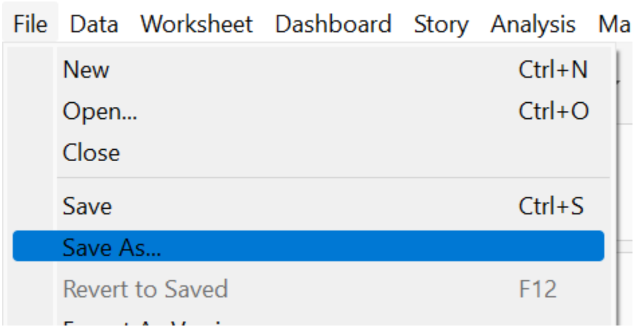






Name the worksheet.

Insights: There were 2 distinct distributions. Temperature < 9, the mode is between 0 – 1.5, happening 24 times. Temperature > 10, the mode is 15 – 16.5, happening 8 times.

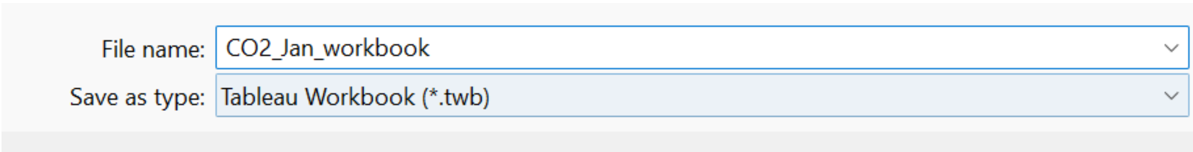


Now, save the workbook.

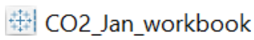
File > Save as

Browse to your folder


Give the filename as "***CO2_Jan_workbook***"



This file will be saved as .twb

Check that the file is properly save in "*MyDvis*".



Is there a rule to set the bin size of a histogram?

No standard rule. You can adjust the bin size until you are able to see some "shape".

Or, you could base on the standard deviation to estimate your bin size. Standard deviation gives you some idea the "spread" interval.

Remember: you need to practice more (go through the steps a few times) to get familiar with the steps.

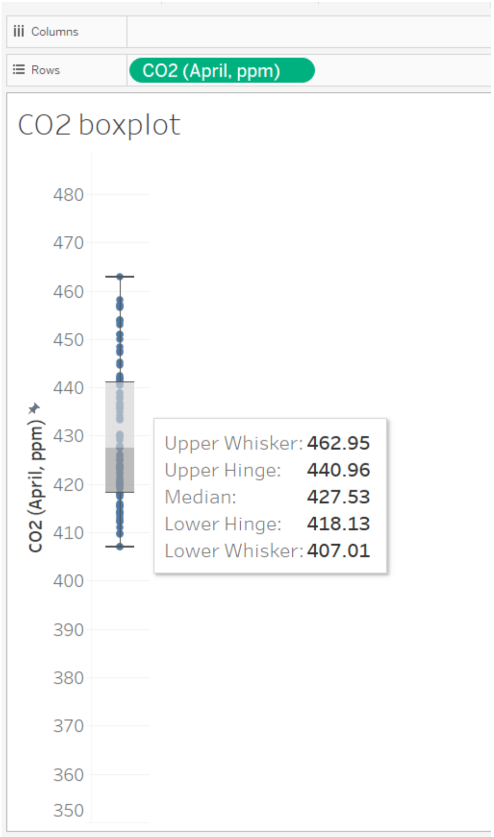
This may feel "tedious" or "boring" but is necessary to train our "mind".

Exercise 9 (Submit to LMS)

Data set: **CO2_data_Apr.xlsx**

a) How many observations were taken? Ans: 97

b) Show a suitable chart to answer the following questions:

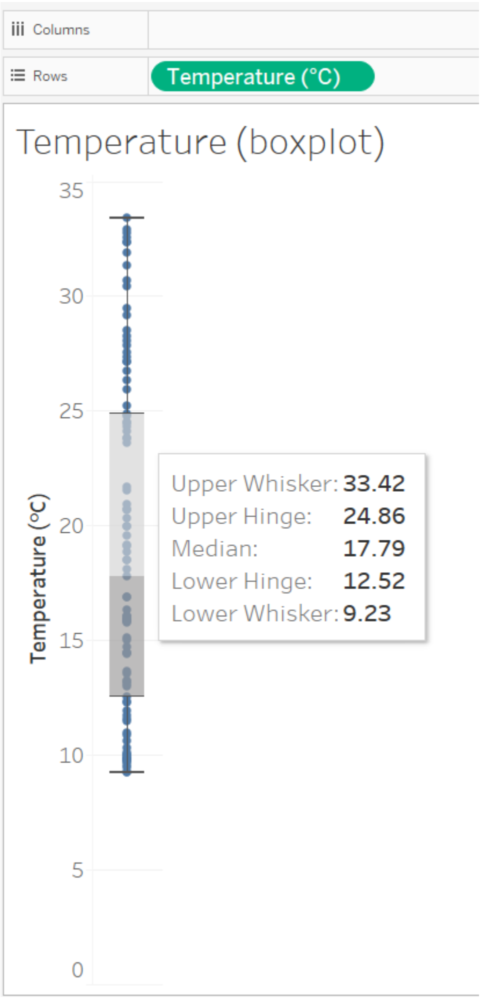


i) What is the range of CO2 level? Ans: 462.95 – 407.01 =55.94

ii) What is the median CO2 in month of Apr? Ans: 427.53

iii) 25% of the data are lower than 418.13

c) Show a suitable chart to answer the following questions regarding the temperature reading:



i) What is the median? Ans: 17.79

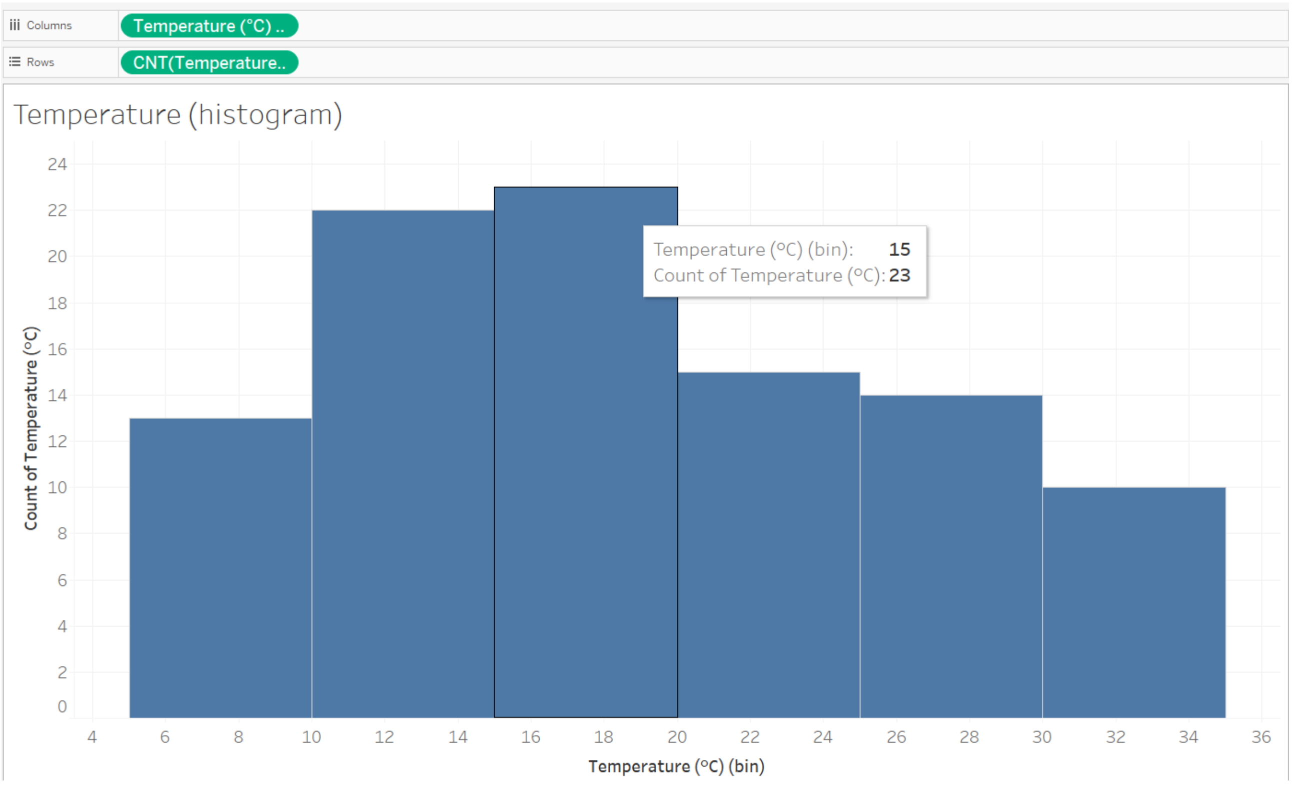
ii) What is the Interquartile range? Ans: 24.86 – 12.52 =12.34

iii) Are there any outliers? Ans: No

iv) Temperature readings _above___ 17.79ºC are widely spread.

Option: (a) above (b) below

4. Create a histogram showing the distribution of temperature, with constant bin interval of 5ºC. Highlight the mode and describe the shape of distribution.



The mode is 15 to 20 ºC.
There is a single mode.
The shape is slightly skewed to the right.

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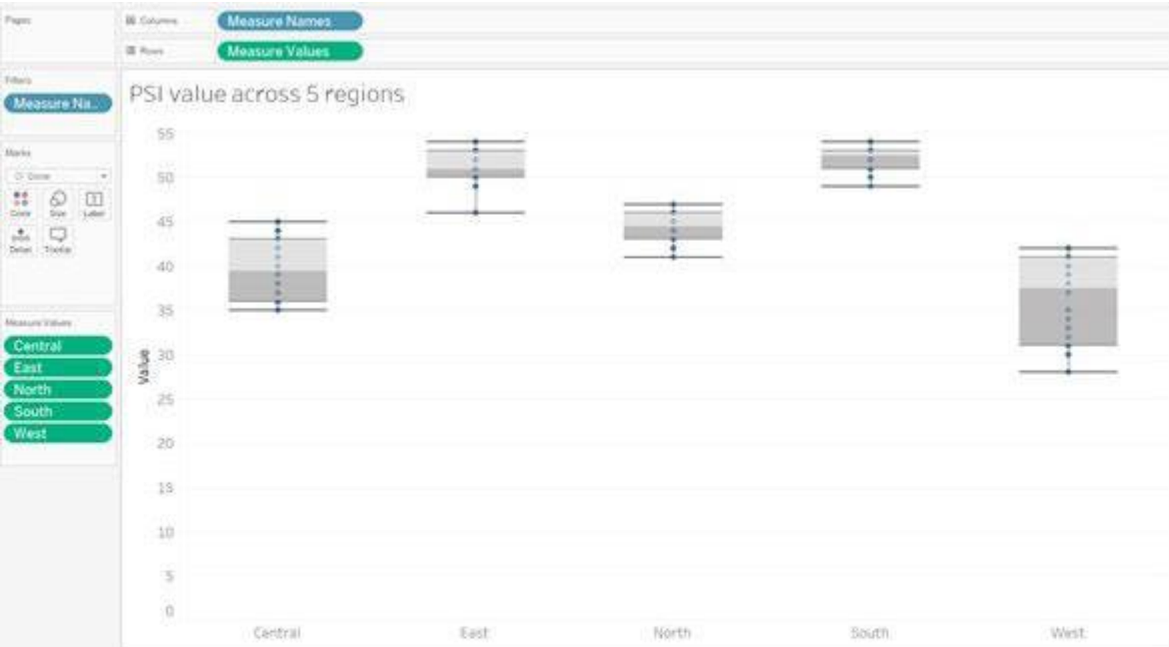
Exercise 10 (Submit to LMS)

Data set: *SG24hrPSI.xlsx*

The data collect the PSI values at 5 regions: North, South, East, West and Central.

Which region has highest Median PSI?

*Support your answer with a suitable visual chart.



South Region has the highest median PSI.